

PRAKTIKUM SISTEM OPERASI

MODUL 4

Membaca Source Code Xinu



Oleh:

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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

DIREKTORAT KAMPUS PURWOKERTO

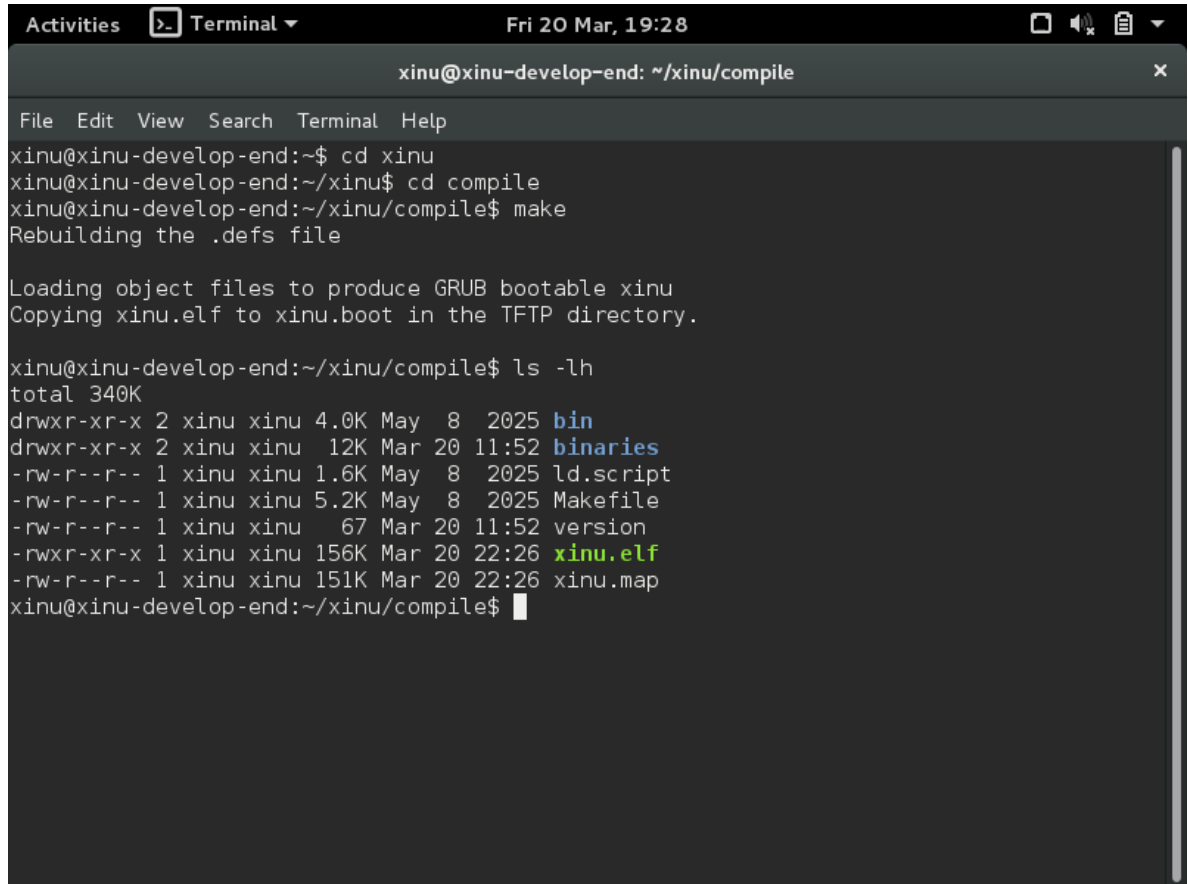
UNIVERSITAS TELKOM

2026

JURNAL/TP

1. **[10 Poin]** Apa nama image yang dihasilkan setelah melakukan kompilasi pada Xinu? Berapa ukuran file tersebut? Ada pada folder apa file image tersebut?

1. Hint: baca kembali modul-modul sebelumnya



```
Activities Terminal Fri 20 Mar, 19:28
xinu@xinu-develop-end: ~/xinu/compile

File Edit View Search Terminal Help
xinu@xinu-develop-end:~$ cd xinu
xinu@xinu-develop-end:~/xinu$ cd compile
xinu@xinu-develop-end:~/xinu/compile$ make
Rebuilding the .defs file

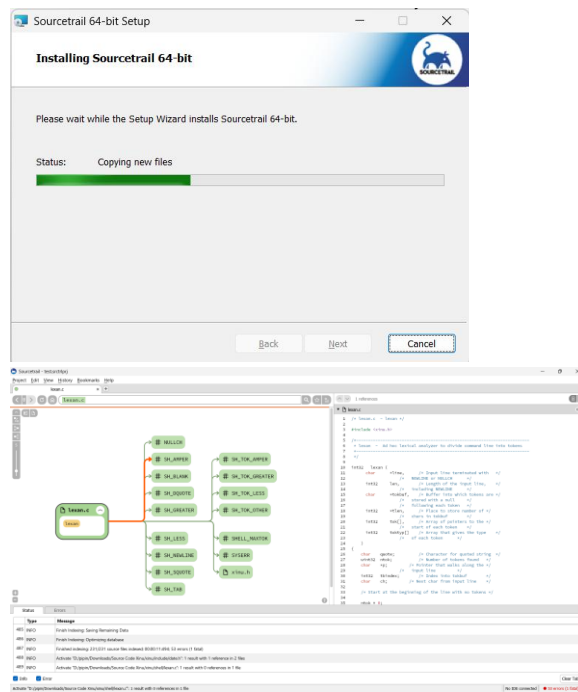
Loading object files to produce GRUB bootable xinu
Copying xinu.elf to xinu.boot in the TFTP directory.

xinu@xinu-develop-end:~/xinu/compile$ ls -lh
total 340K
drwxr-xr-x 2 xinu xinu 4.0K May  8  2025 bin
drwxr-xr-x 2 xinu xinu 12K Mar 20 11:52 binaries
-rw-r--r-- 1 xinu xinu 1.6K May  8  2025 ld.script
-rw-r--r-- 1 xinu xinu 5.2K May  8  2025 Makefile
-rw-r--r-- 1 xinu xinu  67 Mar 20 11:52 version
-rwxr-xr-x 1 xinu xinu 156K Mar 20 22:26 xinu.elf
-rw-r--r-- 1 xinu xinu 151K Mar 20 22:26 xinu.map
xinu@xinu-develop-end:~/xinu/compile$
```

File image yang dihasilkan setelah proses kompilasi Xinu adalah xinu.elf. File tersebut berada pada folder xinu/compile/ dengan ukuran sebesar 156 KB.

2.

- Jawaban:**



3. [10 Poin] Carilah struktur data dari proses pada Xinu OS. Struktur data proses ada pada file apa? Informasi apa saja yang disimpan dalam struktur data tersebut?

Hint: file berektensi .h

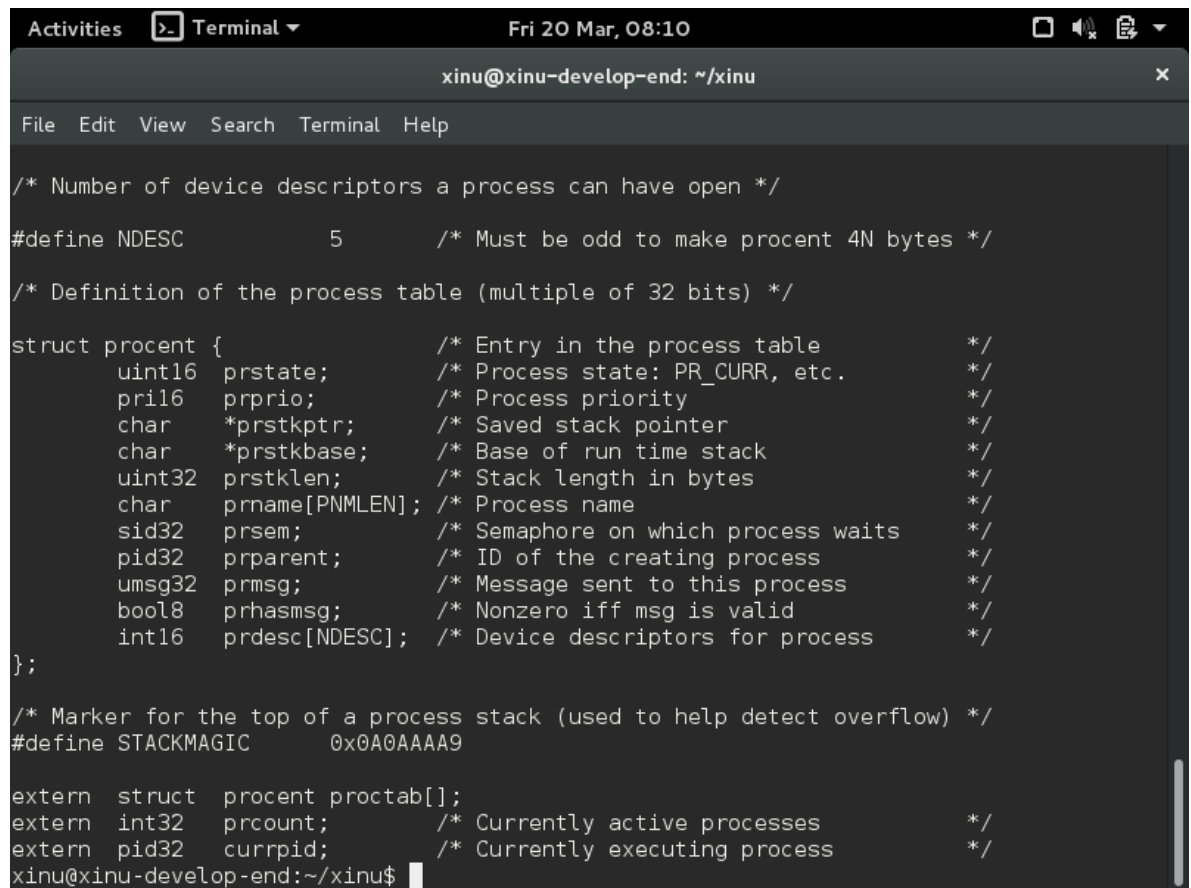
Pada Linux bisa digunakan perintah “grep”

Contoh:

```
$ grep -r "kata_yang_ingin_dicari" /path/ke/direktori/tempat/file/berada
```

3.

Jawab:



```
Activities Terminal Fri 20 Mar, 08:10
xinu@xinu-develop-end: ~/xinu

/* Number of device descriptors a process can have open */
#define NDESC          5          /* Must be odd to make procent 4N bytes */

/* Definition of the process table (multiple of 32 bits) */

struct procent {
    uint16 prstate;          /* Entry in the process table */
    pri16 prprio;            /* Process state: PR_CURR, etc. */
    char *prstkptr;          /* Process priority */
    char *prstkbase;         /* Saved stack pointer */
    uint32 prstklen;         /* Base of run time stack */
    char prname[PNMLEN];     /* Stack length in bytes */
    sid32 prsem;             /* Process name */
    pid32 prparent;          /* Semaphore on which process waits */
    umsg32 prmsg;            /* ID of the creating process */
    bool8 prhasmsg;          /* Message sent to this process */
    int16 prdesc[NDESC];     /* Nonzero iff msg is valid */
};                          /* Device descriptors for process */

/* Marker for the top of a process stack (used to help detect overflow) */
#define STACKMAGIC      0x0A0AAAA9

extern struct procent proctab[];
extern int32 prcount;       /* Currently active processes */
extern pid32 currpri;       /* Currently executing process */
xinu@xinu-develop-end:~/xinu$
```

- a. prstate → status proses (running, ready, dll)
- b. prprio → prioritas proses
- c. prstkptr → pointer ke stack proses
- d. prstkbase → alamat awal stack
- e. prstklen → ukuran/panjang stack
- f. prname → nama proses

- g. prsem → semaphore yang sedang digunakan proses
- h. prparent → ID proses induk (parent)
- i. prmsg → pesan yang dikirim ke proses
- j. prhasmsg → penanda apakah ada pesan atau tidak
- k. prdesc[] → daftar device descriptor yang digunakan proses

4. [80 poin] Mengubah welcome banner pada Xinu

- a. Carilah file yang menyimpan banner Xinu!

Hint: file berekstensi .h pada direktori xinu/include

- b. Carilah file yang menampilkan banner Xinu!

Hint: file berekstensi .c pada direktori xinu/shell

- c. Ubahlah welcome banner Xinu sehingga menjadi seperti ini:

```

Hello World!
Im the first Xinu app and running function main() in system/main.c.
I was created by nulluser() using create().
My creator will turn itself into the do-nothing null process.
I will create a second XINU app that runs shell() in shell/shell.c as an example.
You can do something else or do nothing; it's completely up to you.
...creating a shell

----- NAMA -----

X I N U

----- NIM -----

Welcome to Xinu!

```

Tambahkan NAMA dan NIM masing-masing, kemudian ubah banner Xinu sesuai dengan selera Anda

- d. Setelah melakukan perubahan lakukan perintah berikut ini pada terminal Linux
 - \$ cd xinu/compile
 - \$ make clean
 - \$ make
 - \$ sudo minicom
 - Jalankan vm backend

```
Activities Terminal Fri 20 Mar, 08:14
xinu@xinu-develop-end: ~/xinu

File Edit View Search Terminal Help
xinu@xinu-develop-end:~/xinu$ grep -r "xinu" include/
include/xinu.h:/* xinu.h - include all system header files */
include/date.h:#define ntim2xtim(x) ((x)-2208988800U) /* Net-to-xinu time */
xinu@xinu-develop-end:~/xinu$
```

a.

```
Activities Terminal Fri 20 Mar, 08:33
xinu@xinu-develop-end: ~/xinu

File Edit View Search Terminal Help
GNU nano 2.2.6 File: include/shell.h

/* shell.h - Declarations and constants used by the Xinu shell */

/* Size constants */
#define SHELL_BUFLN TY_IBUFLN+1 /* Length of input buffer */
#define SHELL_MAXTOK 32 /* Maximum tokens per line */
#define SHELL_CMDSTK 8192 /* Size of stack for process */
/* that executes command */
#define SHELL_ARGLEN (SHELL_BUFLN+SHELL_MAXTOK) /* Argument area */
#define SHELL_CMDPRIO 20 /* Process priority for command */

/* Message constants */

/* Shell banner (assumes VT100) */
#define SHELL_BAN0 "\033[31;1m"
#define SHELL_BAN1 "-----"
#define SHELL_BAN2 " "
#define SHELL_BAN3 " X X X X X X X X X X "
#define SHELL_BAN4 " X X X X X X X X X X "
#define SHELL_BAN5 " X X X X X X X X X X "
#define SHELL_BAN6 " X X X X X X X X X X "
#define SHELL_BAN7 "-----"

[ Read 77 lines ]
^G Get Help ^O WriteOut ^R Read File ^Y Prev Page ^K Cut Text ^C Cur Pos
^X Exit ^J Justify ^W Where Is ^V Next Page ^U UnCut Text ^T To Spell
```

b.

```
Activities Terminal Fri 20 Mar, 08:53
xinu@xinu-develop-end: ~/xinu/compile

File Edit View Search Terminal Help

4447448 bytes of free memory. Free list:
[0x0015E320 to 0x0009FFF7]
[0x00100000 to 0x005FBFFF]
115465 bytes of Xinu code.
[0x00100000 to 0x0011C308]
148488 bytes of data.
[0x0011FEC0 to 0x001442C7]

Obt Inteaigned IP address 10.0.3.15 (0x0a00030f)

=====kelvin=====
XINU
=====2311104009=====
Welcome to Xinu!

xsh $ command 6 not found
xsh $ xsh $ xsh $ xsh $ xsh $ xsh $ xsh $ xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Offline | ttyS0
```

c.